

CHAPTER 05

Activities & Lab Packet

Evolution & Natural Selection

A complete set of labs, worksheets, writing prompts, discussions, and a group project to accompany Chapter 5. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S7L5

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1 Bird Beak Buffet Simulation

LAB

OBJECTIVE

Simulate natural selection by testing which "beak" tool best gathers a food source and survives.

TIME

40 min

STANDARDS

S7L5

Materials (per group)

- "Beak" tools: tweezers, spoon, clothespin, chopsticks, binder clip
- "Food": dried beans, rice, gummy worms, marshmallows, toothpicks
- 1 cup per student = the "stomach"
- Timer

Student Instructions

1. Each student picks ONE beak tool — that's your inherited trait. You may not switch.
2. Scatter one food type. On "Go," gather as much as you can into your cup in 30 seconds using only your beak.
3. Count your food. Anyone who caught fewer than 5 pieces "does not survive" this round.
4. Repeat with a different food. Record which beak won each round.

⚠ TEACHER NOTES

Tie it together: no single beak wins every food — the best beak depends on the *environment*. Survivors "reproduce" (their beak type increases). Over many generations the population shifts toward the fittest trait. This is natural selection, not a bird "trying" to change.

Bird Beak Buffet — Data Sheet

Name: _____ Period: _____

Round / Food	My beak tool	Pieces caught	Survived? (Y/N)
1			
2			
3			

Which beak was best for which food? Explain why no beak was best at everything:

2 Camouflage & the Peppered Moth

LAB

OBJECTIVE

Model how a changing environment changes which trait survives, using the peppered moth story.

TIME

30 min

STANDARDS

S7L5

Materials (per pair)

- 1 sheet of light (newspaper/white) "tree bark" and 1 sheet of dark "sooty bark"
- ~30 light paper dots and ~30 dark paper dots (the moths)
- Timer (you are the "predator")

Student Instructions

1. Scatter equal light and dark moths on the LIGHT background. Your partner is the bird: in 10 seconds, grab as many moths as possible (whatever your eye lands on first).
2. Count survivors by color. The survivors "reproduce" — that color increases.
3. Now switch to the DARK (polluted) background and repeat.
4. Record how the surviving color changed when the environment changed.

Peppered Moth — Data Sheet

Name: _____ Period: _____

Background	Light moths eaten	Dark moths eaten	Which color survived better?
Light bark			
Dark (sooty) bark			

During the 1800s, factories darkened the tree bark in England. Use your data to predict what happened to the moth population:

ANSWER KEY

On light bark, light moths blend in and survive; on dark bark, dark moths survive. As pollution darkened the bark, dark moths became far more common — a real, documented case of natural selection driven by an environmental change. When pollution laws cleaned the air, light moths recovered.

3 Fossil Layers & Relative Dating

LAB

OBJECTIVE

Use rock layers to order events in time and infer how organisms changed.

TIME

30 min

STANDARDS

S7L5

Materials (per group)

- The fossil-layer diagram (below) or stacked colored clay with objects pressed in
- Colored pencils

Key Rule

In undisturbed rock, the deepest layer is the oldest and the top layer is the youngest. This is the Law of Superposition.

Student Instructions

1. Study the layers from bottom (oldest) to top (youngest).
2. Number the fossils 1 (oldest) to 5 (youngest).
3. Describe how the organisms appear to change as you move upward.

Fossil Layers — Data Sheet

Name: _____ Period: _____

Layer (top → bottom)	Fossil found	Older or younger than layer below?
Top		
Middle		
Bottom		

How do scientists use the fossil record as evidence that species change over long periods of time?

ANSWER KEY

Deeper layers are older, so fossils lower down lived earlier. By comparing fossils from bottom to top, scientists see gradual changes in body structures over time — strong evidence that species have evolved. Similar fossils in between (transitional forms) connect older and newer species.

4 The Four Steps of Natural Selection

WORKSHEET

OBJECTIVE

Put the steps of natural selection in order and apply them to an example.

TIME

15 min

STANDARDS

S7L5

Name: _____ Period: _____

Part A: Number these steps 1–4 in the correct order.

#	Step
	Helpful traits are passed to offspring, so the trait becomes more common.
	Individuals with traits that fit the environment survive and reproduce more.
	There is natural variation — individuals in a species are not identical.
	Organisms compete because there are limited resources (food, space).

Part B: Apply it. A population of beetles lives on green leaves. Some are green, some are brown.

1. Which beetle has higher **fitness** here, and why? _____
2. After many generations, what color will most beetles be? _____
3. Did individual beetles *change* their own color, or did the *population* change? _____

ANSWER KEY

Part A order: 1) variation, 2) competition for limited resources, 3) survival & reproduction of the fittest, 4) helpful traits passed on / become common. Part B: 1) green beetles — better camouflage from predators; 2) mostly green; 3) the population changed (more green survivors reproduced) — individuals did not repaint themselves.

5 Variation, Adaptation & Fitness

WORKSHEET

OBJECTIVE

Tell the difference between variation, adaptation, and fitness with examples.

TIME

15 min

STANDARDS

S7L5

Name: _____ Period: _____

Part A: Match the term to its meaning.

Term	Meaning
Variation	
Adaptation	
Fitness	
Mutation	

Choices: a trait that helps an organism survive in its environment • differences between individuals of the same species • how well an organism survives and reproduces • a random change in DNA that can create new variation

Part B: Label each as a Physical (structural) or Behavioral adaptation.

Adaptation	Physical / Behavioral
A cactus has spines instead of leaves	
Birds fly south for the winter	
An arctic fox grows thick white fur	
Wolves hunt in packs	

Part C:

1. Where does the NEW variation that natural selection acts on originally come from? _____
2. Does "fitness" mean the strongest animal? Explain. _____

ANSWER KEY

Part A: Variation = differences between individuals; Adaptation = a trait that aids survival; Fitness = how well an organism survives and reproduces; Mutation = a random DNA change creating new variation. Part B: spines = physical; migration = behavioral; thick fur = physical; pack hunting = behavioral. Part C: 1) mutations (random DNA changes); 2) no — fitness means best at surviving AND reproducing in that environment, not necessarily strongest.

6 Reading the Fossil Record

WORKSHEET

OBJECTIVE

Identify the kinds of evidence scientists use to support evolution.

TIME

20 min

STANDARDS

S7L5

Name: _____ Period: _____

Part A: Name the type of evidence each describes (fossil record, similar structures, DNA, or embryos).

Evidence	Type
A whale flipper and a human arm have the same bone pattern	
Deeper rock layers hold older, simpler organisms	
Closely related species share more genetic code	
Many animal embryos look alike in early development	

Part B: Short answer.

1. What is a transitional fossil, and why is it useful evidence? _____
2. Why are most organisms NOT preserved as fossils? _____
3. Two animals share very similar DNA. What does that suggest about their history? _____

ANSWER KEY

Part A: whale/human bones = similar (homologous) structures; rock layers = fossil record; shared genetic code = DNA; embryos = embryology. Part B: 1) a fossil showing features between an older and newer group — it links them and shows gradual change; 2) preservation is rare — soft bodies decay and conditions must be just right; 3) they likely share a recent common ancestor.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain evolution and natural selection in writing.

TIME

15–30 min each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Built to Survive.** Pick an animal and describe one adaptation that helps it live in its environment. Use the words trait and survive.
2. **Same but Different.** Explain what variation means and give an example from your own classroom.

Level 2 — Intermediate (3 paragraphs)

3. **The Story of the Moths.** Explain how the peppered moth population changed when the environment changed. Walk through all four steps of natural selection.
4. **A Common Mistake.** Some people think animals "try" to evolve or change their bodies on purpose. Explain why that is incorrect and what really happens.

Level 3 — Advanced (essay)

5. **Weight of the Evidence.** Scientists support evolution with several independent lines of evidence (fossils, similar structures, DNA, embryos). Construct an argument explaining why having MANY different kinds of evidence makes a scientific explanation stronger than just one.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific reasoning through think-pair-share.

TIME

Warm-ups, exit tickets, seminar

STANDARDS

Cross-cutting

TEACHER NOTES

Think-pair-share format. Several questions have no single right answer — push for evidence from the chapter.

Quick Warm-Ups (1–2 min)

1. Does an individual organism evolve in its lifetime, or does a population evolve over generations?
2. Why is variation within a species so important for survival?
3. If a trait helps survival but the animal never reproduces, does it increase in the population? Why not?
4. Where do brand-new traits originally come from?
5. Name an adaptation that is behavioral, not physical.

Deeper Discussions (5–15 min)

6. How are natural selection and selective breeding similar, and what is the key difference between them?
7. A fast-spreading bacteria becomes resistant to a medicine. Explain how natural selection caused this.
8. Why might a trait that is helpful in one environment be harmful in another? Give an example.
9. Why does it take MANY generations for a population to change noticeably?
10. Some species alive today look almost identical to ancient fossils. Does that disprove evolution? Explain.

9 Build-a-Survivor Design Challenge

GROUP

OBJECTIVE

Design an organism whose adaptations let it survive a specific environment.

TIME

2 class periods

STANDARDS

S7L5

Project Description

Each group is given (or draws) an extreme environment. They invent an organism perfectly adapted to it, then explain how natural selection could have produced those adaptations. Present as a labeled poster or model.

Project Requirements

The Environment

- Climate & key abiotic factors
- Available food and main predators
- One major survival challenge

The Organism

- ≥ 3 physical adaptations (labeled)
- ≥ 1 behavioral adaptation
- A short "natural selection story" explaining how the adaptations became common

Presentation Day

Groups present their survivor. The class identifies each adaptation and the survival problem it solves. Bonus: predict what would happen to the organism if its environment suddenly changed.

! TEACHER NOTES

Environment ideas: deep ocean trench, scorching desert, frozen tundra, dense rainforest canopy, a cave with no light. **Grading:** adaptations fit the environment 40%, correct natural-selection reasoning 30%, creativity 15%, presentation 15%.

10 Vocabulary Bingo

REVIEW

OBJECTIVE

Review chapter vocabulary in a low-stress game.

TIME

20–30 min

STANDARDS

S7L5

Setup

Blank 5×5 grid; word list on board; FREE middle. Read a clue, students mark the word. 5 in a row wins.

Vocabulary List (24 words)

evolution	natural selection	adaptation	variation	trait
fitness	mutation	species	population	environment
survival	reproduce	inherited	camouflage	predator
prey	competition	fossil	fossil record	extinct
common ancestor	selective breeding	generation	resistance	—

Sample Clue Cards (read aloud, don't show)

Clue to read aloud	Answer
The change in a species' traits over many generations	evolution
Survival of individuals whose traits fit the environment	natural selection
A trait that helps an organism survive in its surroundings	adaptation
Differences between individuals of the same species	variation
How well an organism survives AND reproduces	fitness
A random change in DNA that can create a new trait	mutation
Coloring or patterns that help an animal blend in	camouflage
When the last members of a species die out forever	extinct
Preserved remains or traces of ancient organisms	fossil
An ancestor shared by two or more different species	common ancestor
When organisms struggle over limited food or space	competition
When humans choose which organisms reproduce	selective breeding

! TEACHER NOTES

Easy adaptation: give the list with definitions during play. Run multiple patterns so several students win.